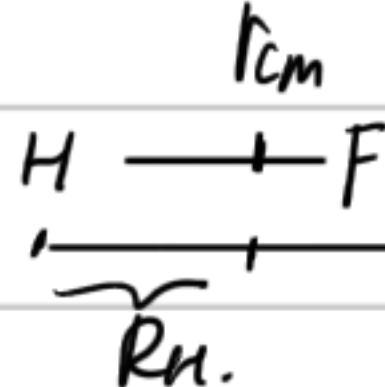


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P1. $m_H R_H^2 + m_F R_F^2 = I$

set



as the coordinate.

so $r_{cm} = R_H$

$(R_H + R_F) m_F = (m_H + m_F) R_H$

$R_F = 4.65 \times 10^{-12} \text{ m}$

$R_H = 8.82 \times 10^{-11} \text{ m}$

$r = R_F + R_H = 9.29 \times 10^{-11} \text{ m}$

P2. to prove $I_z = I_x + I_y$ where

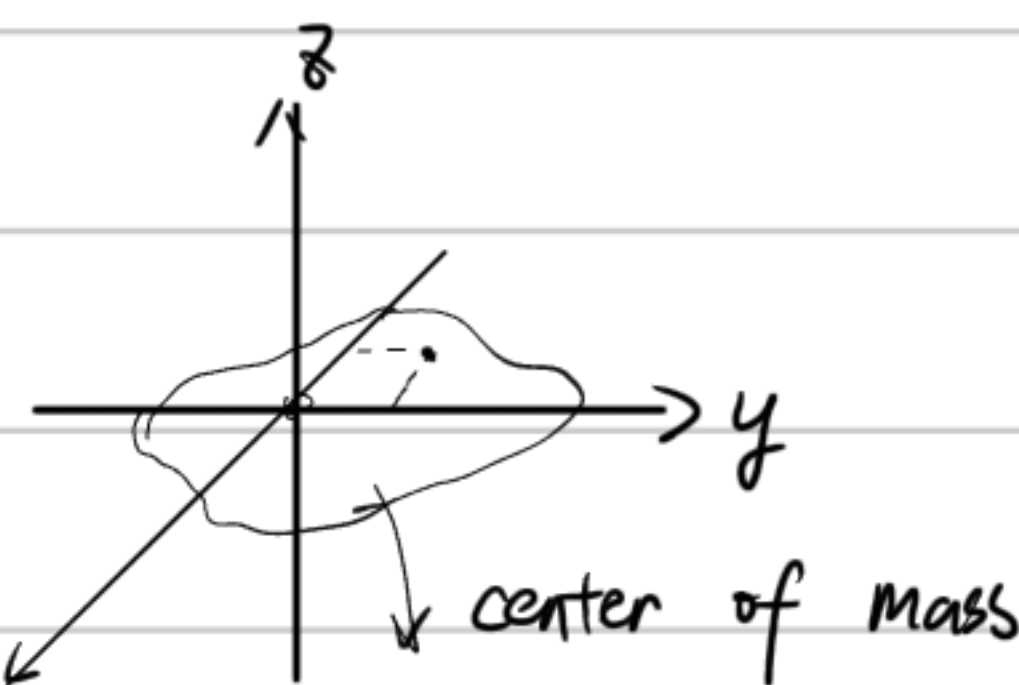
since $I_z = \int r_z^2 dm$

for any point on the plane with mass to be dm

$r_z^2 = r_x^2 + r_y^2$

so $\int r_z^2 dm = \int r_x^2 dm + \int r_y^2 dm$

$\Rightarrow I_z = I_x + I_y$



P3 for a slice of the hollow cylinder

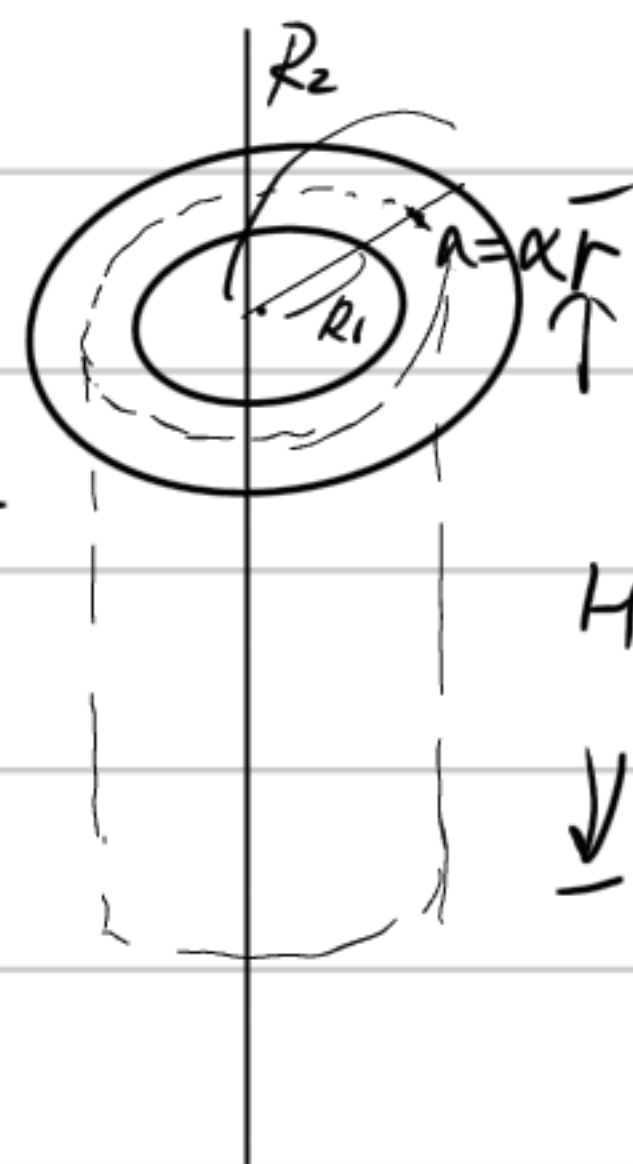
$I = \int r^2 dm$ $dm = 2\pi r H dr \cdot \alpha$

$I = \int r^2 \cdot 2\pi r H \alpha dr$

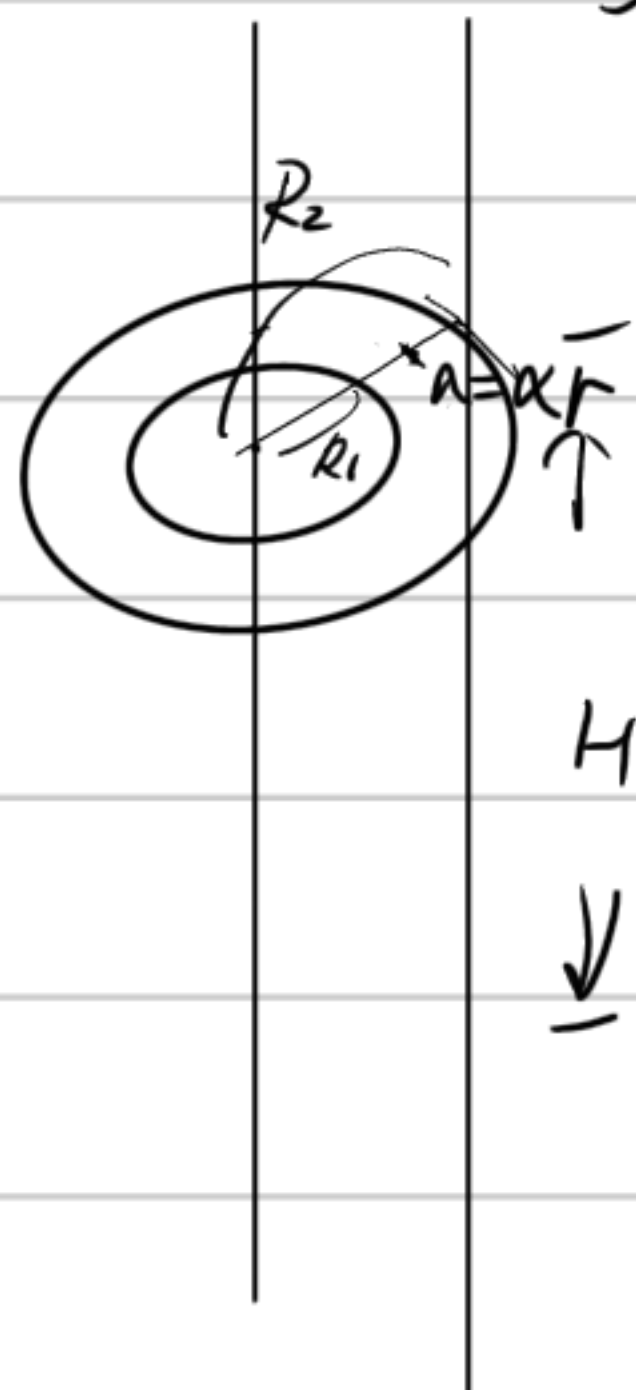
$= \int_{R_1}^{R_2} 2\pi H \alpha r^3 dr$

$= \frac{2}{5} \pi H \alpha r^5 \Big|_{r=R_1}^{r=R_2}$

$= \frac{2}{5} \pi H \alpha (R_2^5 - R_1^5)$



P4.

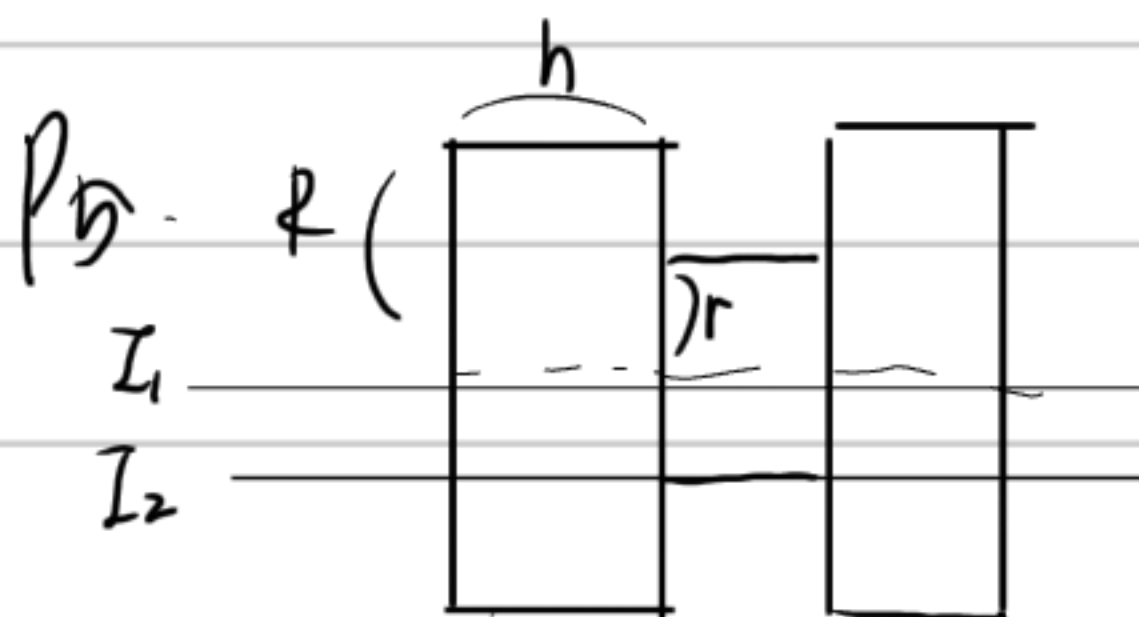


$I_t = I + mb^2$

$b = R_2$

$m = \int dm = \int_{R_1}^{R_2} 2\pi H \alpha r^2 dr = \frac{2}{3} \pi H \alpha (R_2^3 - R_1^3)$

so $I_t = I + mb^2 = \frac{2}{5} \pi H \alpha (R_2^5 - R_1^5) + \frac{2}{3} \pi H \alpha (R_2^3 - R_1^3) R_2^2$
 $= \frac{16}{15} \pi H \alpha R_2^5 - \frac{2}{5} \pi H \alpha R_1^5 - \frac{2}{3} \pi H \alpha R_1^3 R_2^2$



the masses : $\pi R_1^2 h \alpha$, $\pi r^2 h \alpha$, $\pi R_2^2 h \alpha$ $m_1 = m_3$

set I with axis pass through center of mass

$I = \frac{1}{2} m_1 R^2 + \frac{1}{2} m_2 r^2 + \frac{1}{2} m_3 R^2 = m_1 R^2 + \frac{1}{2} m_2 r^2$

$$\text{so } I_2 = I + (m_1 + m_2 + m_3) r^2 = m_1 R^2 + \frac{1}{2} m_2 r^2 + (2m_1 + m_2) r^2$$

$$= m_1 R^2 + \frac{3}{2} m_2 r^2 + 2m_1 r^2 \quad m_2 = \frac{r^2}{R^2} m_1$$

when it's falling along the string



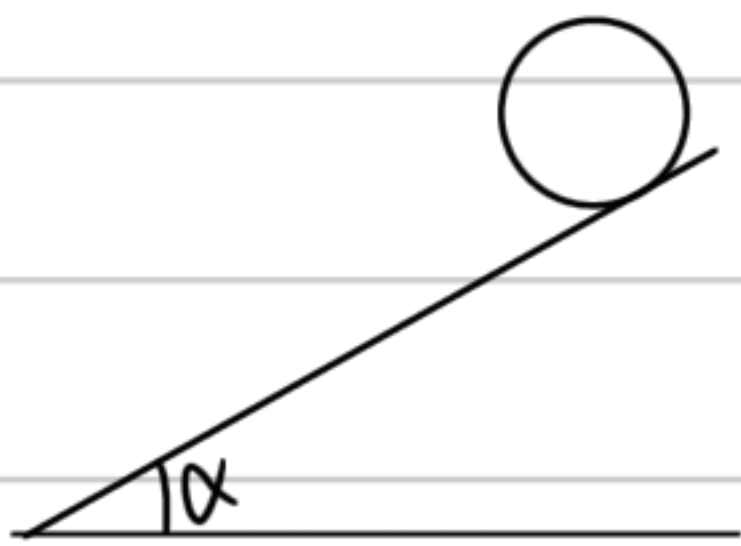
$$Mg r = I_2 \epsilon$$

$$a_{cm} = r \epsilon$$

$$a_{cm} = \frac{(4R^2 + 2r^2) g r^2}{2R^4 + 3r^4 + 4R^2 r^2}$$

direction: downwards

P6



for the ball:

$$I_b = \frac{2}{5} m R^2 + m R^2 = \frac{7}{5} m R^2$$

$$R m g \sin \alpha = I_b \epsilon$$

$$a_{cm} = R \epsilon$$

$$\epsilon = \frac{R m g \sin \alpha}{\frac{7}{5} m R^2} = \frac{5}{7} \frac{g \sin \alpha}{R}$$

$$a_{cm} = \frac{5}{7} g \sin \alpha$$

for the ring

$$I' = m' R'^2 + m' R'^2 = 2m' R'^2$$

$$R' m' g \sin \alpha = I' \epsilon'$$

$$a_{cm}' = R' \epsilon'$$

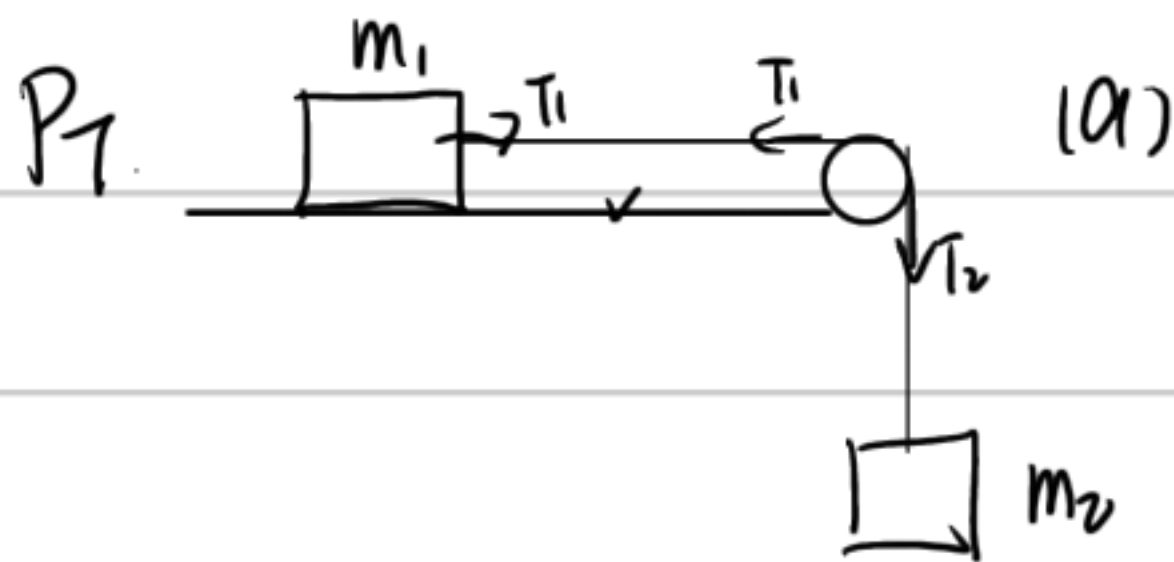
$$\epsilon' = \frac{R' m' g \sin \alpha}{2 m' R'^2} = \frac{g \sin \alpha}{2 R'}$$

$$a_{cm}' = \frac{1}{2} g \sin \alpha$$

$$\frac{1}{2} a_{cm} t^2 = v_0 t + \frac{1}{2} a_{cm}' t^2$$

$$v_0 = \frac{3}{28} g \sin \alpha t$$

direction: parallel to the inclined plane



(a)

$$\begin{cases} R(T_2 - T_1) = I \epsilon \\ I = \frac{1}{2} M R^2 \\ a = R \epsilon \\ T_1 = m_1 a \\ m_2 g - T_2 = m_2 a \end{cases}$$

$$\epsilon = \frac{2m_2 g}{2m_1 R + 2m_2 R + M R}$$

$$\Rightarrow a = \frac{2m_2 g R}{2m_1 R + 2m_2 R + M R}$$

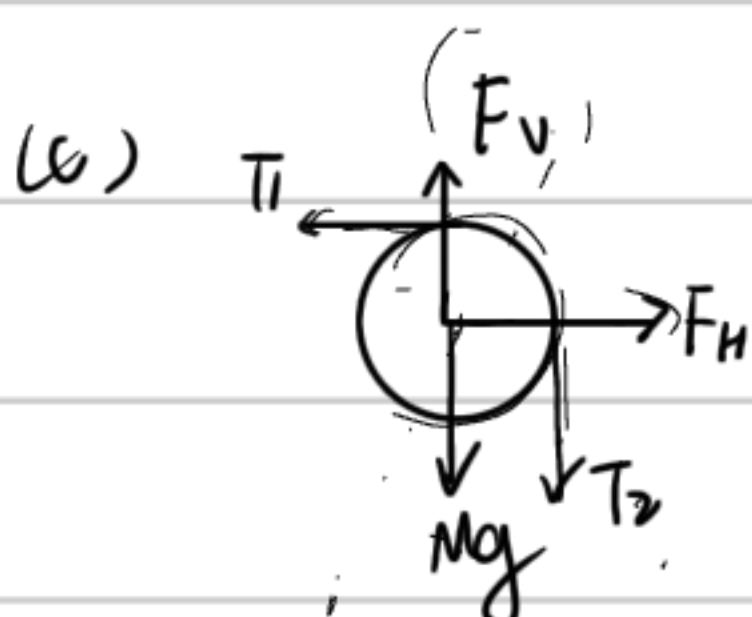
$$T_1 = \frac{2m_1 m_2 g}{2m_1 + 2m_2 + M}$$

$$T_2 = \frac{2m_1 m_2 g + m_2 M g}{2m_1 + 2m_2 + M}$$

(b) $a = \frac{2m_2 g}{2m_1 + 2m_2 + M}$

for m_1 its direction is right

for m_2 its direction is downwards

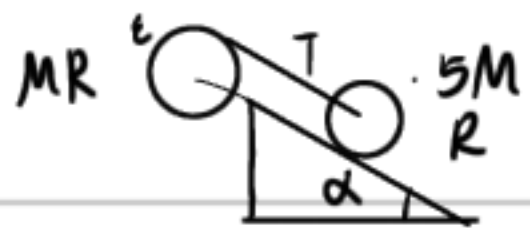


$$F_v = Mg + T_2 = Mg + \frac{2m_1 m_2 g + m_2 M g}{2m_1 + 2m_2 + M}$$

upwards

$$F_H = T_1 = \frac{2m_1 m_2 g}{2m_1 + 2m_2 + M}$$

horizontal



P8. $I_b = \frac{2}{5} \times 5MR^2 + 5MR^2$ $I_b = 7MR^2$

$$5Mg \sin \alpha - T/R = I_b \epsilon$$

$$a_{cm} = R\epsilon$$

$$TR = \frac{1}{2}MR^2 \cdot \epsilon$$

$$\epsilon = \frac{2}{3} \frac{g \sin \alpha}{R}$$

$$a_{cm} = \frac{2}{3} g \sin \alpha \quad \text{downwards, parallel to the inclined plane}$$

x is the deviation distance from equilibrium point

P9. $mg - T_1 = ma$

$$a = \epsilon R$$

$$(T_1 - T_2)R = I\epsilon$$

$$T_2 = k(x + x_0)$$

$$\Rightarrow \ddot{x} + \frac{kR^2}{MR^2 + I} x = 0$$

$$\omega_0^2 = \frac{kR^2}{MR^2 + I}$$

$$\omega_0 = \sqrt{\frac{kR^2}{MR^2 + I}}$$

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{R} \sqrt{\frac{MR^2 + I}{k}}$$